Lab 3 — Common Issues

ECON 470 — Spring 2026

A handful of patterns came up across many of your submissions. None of them is fatal on its own, but together they account for most of the points the class lost. Read these alongside the answer key.

Part B(d) and C(c): plug-back-in arithmetic

Three groups got $\Delta^*_{DEC} = 25$ in Part B and $\Delta^*_{DEC} = 15$ in Part C correct — and then made arithmetic slips evaluating π^*_{DEC}. The correct calculations are short:

$$\pi^*_{DEC}(B) = 75 + 0.25(25) - 0.005(25)^2 - 5 = 75 + 6.25 - 3.125 - 5 = 78.125$$

$$\pi^*_{DEC}(C) = 0.85(100) + 0.15(35) - 0.005(15)^2 - 5 = 85 + 5.25 - 1.125 - 5 = 84.125$$

The slips were in the middle term: $0.25 \times 25 = 6.25$ (not 13.75); $0.85(100) + 5.25 - 1.125 - 5 = 84.125$ (not 85.25). After the slip, the *qualitative* answer (adopt vs. don't, comparison to the rule payoff) usually still came out right — but the gain or comparison delta was off by 1-2 points and the rest of the analysis inherited the wrong number. A 30-second sanity check before moving on catches this kind of error: the rule payoff is 83 and AI helps; the gain in Part C is small (about 1.1), not 2 or 3. If you're seeing 85+ for $\pi^*_{DEC}(C)$, recompute.

Part D(b): "high e means less insurance" is the conclusion, not the argument

Several answers reasoned: *"when e is high, the firm makes fewer mismatches, so it needs less insurance."* That's the right intuition for *why* insurance falls under a decision — but it isn't the question. The question asks why *the same threshold* $e = p + x^*_{RULE}$ separates both regions. Answering it requires a small amount of algebra:

Under a decision, the FOC $1 - e = \kappa\Delta^*_{DEC}$ gives $\Delta^*_{DEC} = (1 - e)/\kappa$.

Under a rule, the FOC $1 - p - x^* = \kappa\Delta^*_{RULE}$ gives $\Delta^*_{RULE} = (1 - p - x^*)/\kappa$.

Insurance rises iff $\Delta^*_{DEC} > \Delta^*_{RULE}$, which (after dividing by $\kappa > 0$) is $1 - e > 1 - p - x^*$, which is $e < p + x^*_{RULE}$. The same inequality determines adoption (Problem 2's argument). The two thresholds aren't *similar* — they're *the same threshold*, and showing that is one line of algebra.

Part D(c): the question is "what is Prop 4.1 actually doing?"

A few answers pivoted to a discussion of *when* the firm adopts AI ("the necessary condition is $e > p$" or "really the firm cares about $e > p + x^*$"). That's an answer to a different question. D(c) asks: given that a freely-optimizing firm will never actually experience rising insurance, *what is Proposition 4.1 telling us?*

The cleanest framing is: it's a *comparative-statics* statement, not a positive prediction. It compares two optimization problems — "what would you want under a rule?" vs. "what would you want under a decision?" — without claiming the firm faces both. That makes it useful in *forced-regime* settings: if regulation, a competitive mandate, or an organizational policy pushes the firm into the decision regime at a signal quality where it would not have voluntarily adopted, Prop 4.1 tells us insurance spending will rise. The proposition's value is *counterfactual analysis*, which is exactly what comparative statics is for.

Part A(b) and A(c): name the marginal benefit and the marginal cost

Q(b) asks for the marginal benefit and the marginal cost of protection. The full answer names both, in terms of the model:

- **Marginal benefit:** $R - r - \Delta$. An additional unit of protection shifts probability to the matching state, raising the realized payoff from $r + \Delta$ to R. The gap $R - r - \Delta$ is the per-unit gain.
- **Marginal cost:** $\gamma x = 600x$. The cost function is quadratic, so the marginal cost rises linearly in x.

A few answers said "the marginal cost is the derivative of the cost function" or "the cost is increasing." That's true but not specific — the question wants the *expression* γx. The same applies in (c) for $\kappa\Delta$ as the marginal cost of insurance.

Part A(d): show the substitution

A handful of answers reported $x^* = 0.1$ and $\Delta^* = 20$ without showing how. The system of two FOCs is

$$80 - \Delta = 600x \quad \text{and} \quad 0.3 - x = 0.01\Delta.$$

From the second, $\Delta = 100(0.3 - x) = 30 - 100x$. Substitute into the first:

$$80 - (30 - 100x) = 600x \implies 50 + 100x = 600x \implies x^* = 0.1.$$

Three lines. The point of showing work isn't pedantry — it's that if your final answer is wrong, you can trace where the error entered. Bare numerical answers leave no trail.

A note on the "blank handout" submission

One submission this round was the canvas handout itself, with no answers added. The grading workflow checks for this by hashing each submission against the original handout, but the file in question had a slightly different MD5 (different export of the same content), so the check missed it. From a workflow perspective, hashing is a fast first pass; *reading the body* is what catches re-uploaded handouts that have been re-exported. From your perspective: the simplest way to avoid this category of mistake is to skim the rendered PDF before clicking submit, just to confirm your answers actually made it through.

A meta-comment on showing work and units

Several technically-correct submissions lost a few points for the same reason: brief answers that left the *connection* between steps invisible. Lab 3 has a particular structure — every Part-B and Part-C numerical answer should explicitly cite the rule benchmark ($\pi^*_{RULE} = 83$, $\Delta^*_{RULE} = 20$) when comparing. The cleanest answers said things like "$\pi^*_{DEC} = 84.125 > 83 = \pi^*_{RULE}$, so the firm adopts." That phrasing makes the comparison machinery visible. "$84.125 > 83$, so adopt" is correct but harder to grade because the reader has to remember which number is which.

The two cleanest submissions in the class did this consistently — every numerical comparison was framed against its named benchmark, and the algebra in Part D(b) was written out rather than gestured at. That's what turns a writeup of *answers* into an *argument*.